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RALPH O'CONNOR SUSTAINABLE ENERGY INSTITUTE

When the Supply Chain Shapes the Grid:
Integrating Upstream Constraints into Long-Term Planning

Boyu Yao
10/07/2025

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Electric Power Innovation for a Carbon-free Society (EPICS)
Academic center for Reliability and Resilience of Offshore Wind (ARROW)
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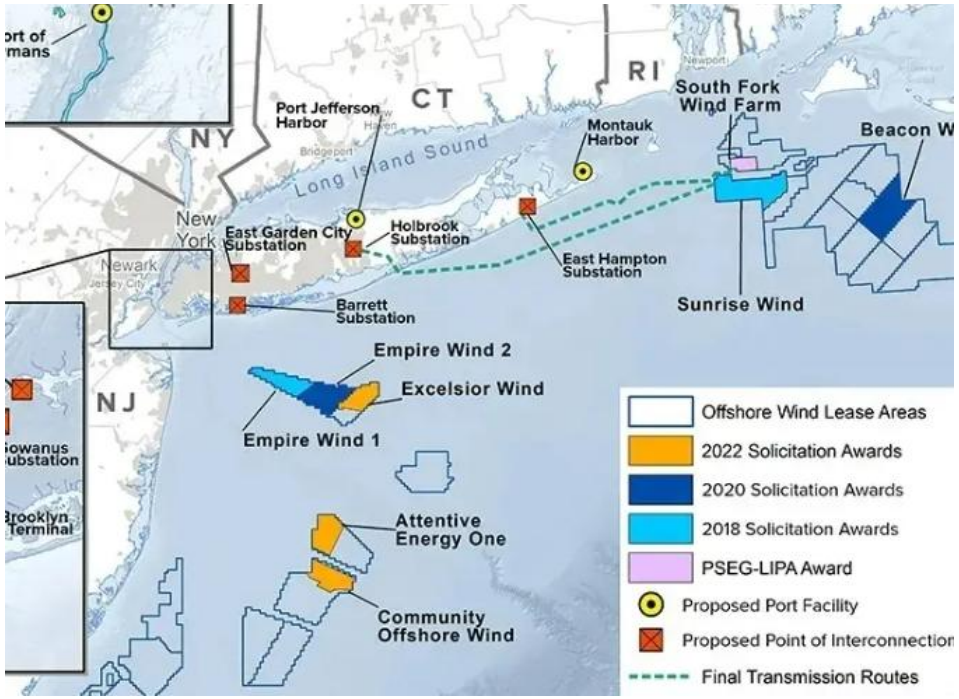
Department of Civil and Systems Engineering, Johns Hopkins University

Energy at Hopkins

2 Stories

- Story 1 — The Broken Promise (New York Wind)
  - **2021:** New York awarded contracts for massive offshore wind farms (e.g., *Empire Wind 2*), meant to power millions of homes [1].
  - Contracts locked in, market signals clear, developers ready to build.
  - **By 2024:** plans collapsed — inflation, material bottlenecks, and shortages of submarine cables and turbines [2].
  - The contract was torn up: the market promised clean megawatts, but the supply chain couldn't deliver.

Supply chain — not contracts — dictated what was feasible.



Sif Removes Empire Wind 2 from Order Book

FOUNDATIONS

January 4, 2

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Empire Wind 2 Partners Cancel USD 250 Million Offshore Substation Contract

CONTRACTS & TENDERS

January 4, 2024, by A

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JANUARY 5, 2024

Offshore wind contracts with Empire Wind 2 cancelled

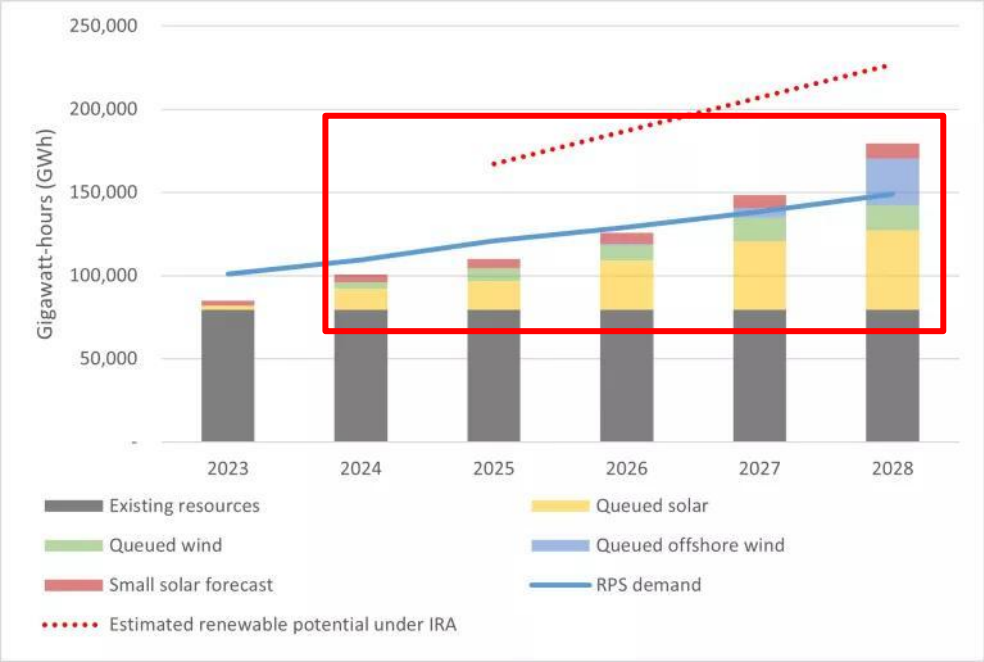
by WorkBoat Staff in Wind, Offshore, News

SHARE

[1] https://www.nyserda.ny.gov/All-Programs/Offshore-Wind/Focus-Areas/NY-Offshore-Wind-Projects
[2] https://patch.com/new-jersey/longbranch/empire-wind-2-turbines-canceled-would-have-been-19-miles-nj

2 Stories

- Story 2 — The Price Shock (PJM Capacity Market)
  - **2024:** PJM capacity auctions saw prices spike up to **800%** in some zones [3].
  - Consumers and businesses faced billions in extra costs.
  - **"We need new capacity, now!"**
  - Projects stalled in queues, with transformers, materials, and financing all in short supply.



Supply chain — not planning models — determined which projects could proceed.

[3] https://www.reuters.com/business/energy/pjm-power-auction-results-yield-sharply-higher-prices-2024-07-31
[4] https://opc-dc.gov/wp-content/uploads/2025/05/PJM-Capacity-Market-Report-FINAL-OPC-Synapse.pdf

Why? Mismatch between planning and supply chain reality!

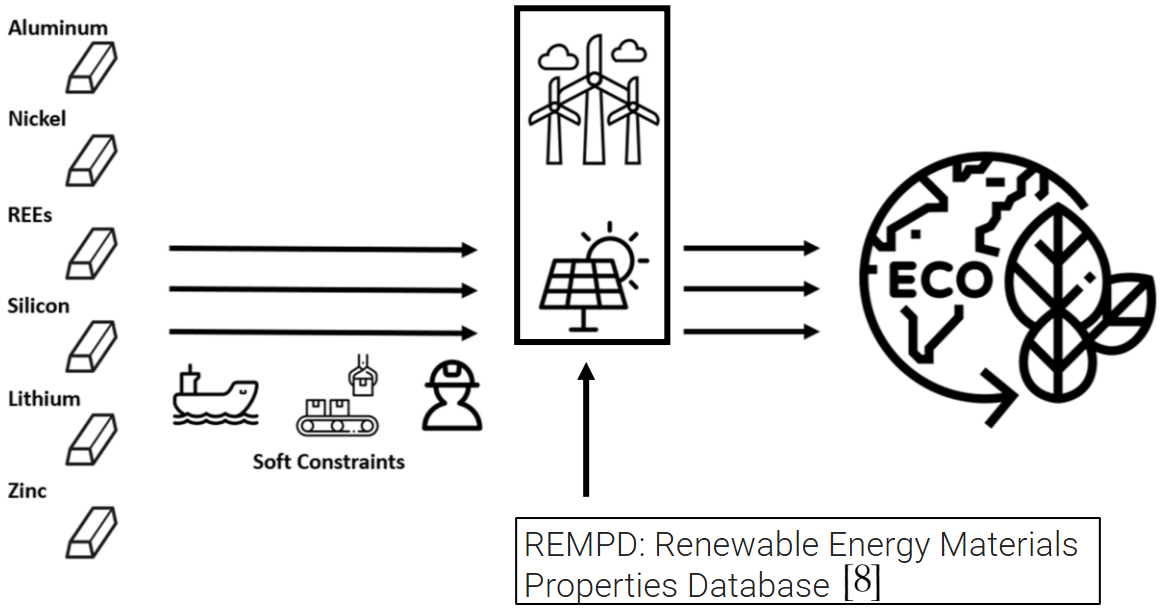
- Historical Stability
  - Traditional assets (coal, gas, nuclear) relied on abundant, low-cost materials (steel, cement).
  - Supply costs were <5% of capex, even during price spikes [5]
  - Strong, guaranteed returns reinforced perception that upstream constraints weren't limiting [6].
  - Shift only recent: wind, solar, and storage depend on global, modular, supply-intensive chains.

- Data & Uncertainty
  - Downstream data (load, generation, prices) well-documented.
  - Upstream data (material availability, lead times) scarce & uncertain.
  - Lack of standardized datasets hinders formal modeling of supply chains.

- Modeling Gaps
  - Existing tools siloed: IAMs → global scale, coarse, long-term. LCAs → retrospective, static. Traditional GEP → ignores upstream.
  - No unified framework embedding supply dynamics into operationally relevant planning.

[5] Rong, F., & Victor, D. G. (2012). What does it cost to build a power plant. Laboratory on International Law and Regulation ILAR.
[6] IEA (2024), Accelerating Just Transitions for the Coal Sector, IEA, Paris https://www.iea.org/reports/accelerating-just-transitions-for-the-coal-sector, Licence: CC BY 4.0

A decision framework for clean energy supply chain [7]



REMPD: Renewable Energy Materials Properties Database [8]

Material intensity per unit capacity (e.g., tons/MW for benchmark plants)

[7] Yao, B., & Dvorkin, Y. (2025, July). A decision framework for clean energy supply chain optimization. In Proceedings of the IEEE Power & Energy Society General Meeting (PESGM). IEEE. (Forthcoming)
[8] NREL (National Renewable Energy Laboratory). 2023. "Renewable Energy Materials Properties Database." Version 1.1. Washington, D.C.: U.S. Department of Energy. https://apps.openei.org/REMPD/.

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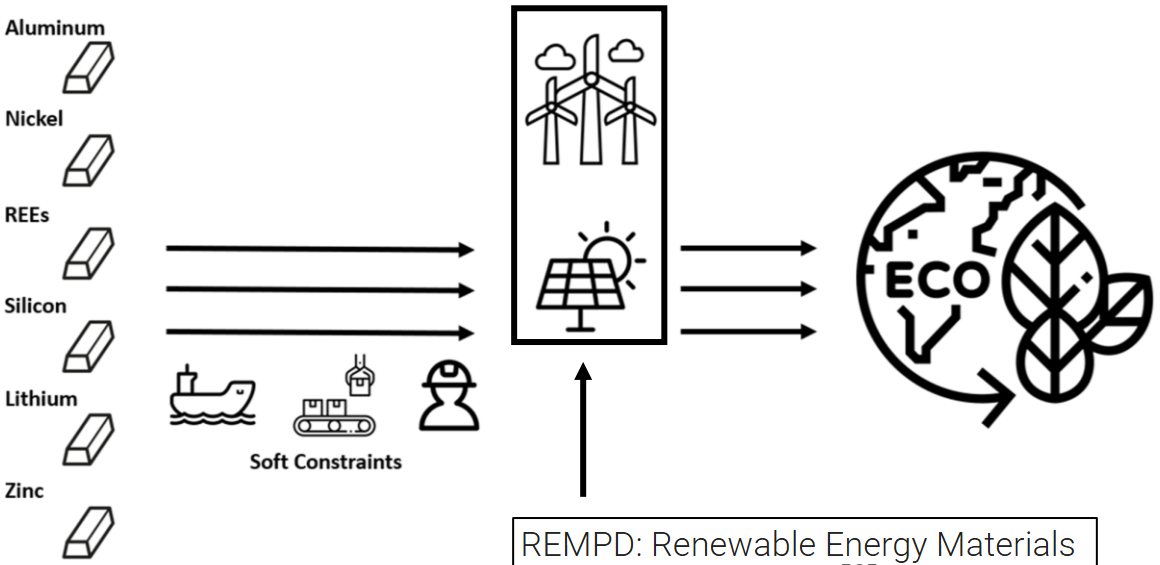
14 critical materials from the overlap of the USGS and DOE critical materials lists [9, 10].

Project future material availability based on the last supercycle.



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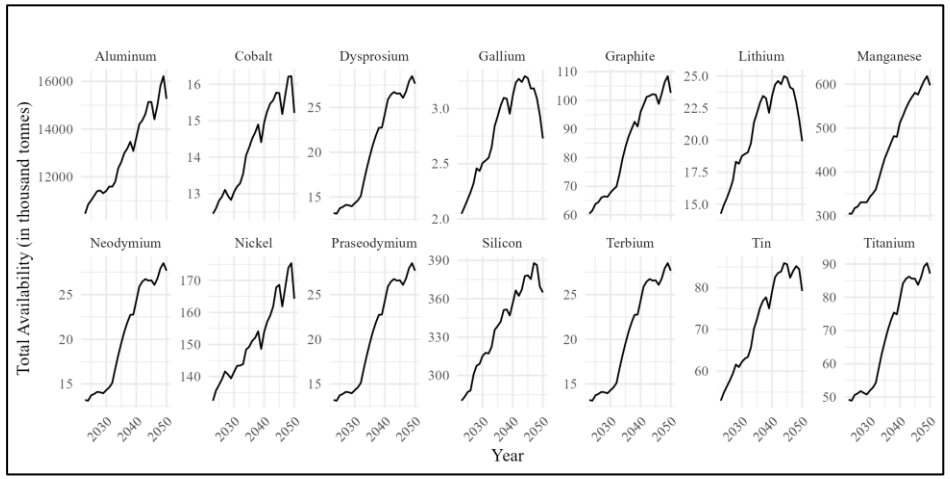

$$(1)\quad \min \sum_{t\in\mathcal{T}} \frac{1}{(1+D)^{t-1}} \left(\sum_{m\in\mathcal{M}} C^{\mathrm{rm}}_{m,t} \cdot \left(\sum_{o\in\mathcal{I}} p_{o,m,t} + \sum_{o\in\mathcal{D}} \frac{q_{o,m,t}}{R^{\mathrm{c}}_{o,m,t}} \right) + \sum_{p\in\mathcal{P}} C^{\mathrm{p}}_{p,t} \cdot x_{p,t} \right)$$

$$(2)\quad y_{m,t} \geq \sum_{c\in\mathcal{C}} z_{c,t} \cdot D^{\mathrm{c}}_{m,c}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(3)\quad z_{c,t} \geq \sum_{p\in\mathcal{P}} x_{p,t} \cdot D^{\mathrm{p}}_{c,p}, \quad \forall c \in \mathcal{C},\ t \in \mathcal{T}$$

$$(4)\quad p_{o,m,t} \leq Y_{o,m,t}, \quad \forall o \in \mathcal{I},\ m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(5)\quad q_{o,m,t} \leq R^{\mathrm{c}}_{o,m,t} \cdot Y_{o,m,t}, \quad \forall o \in \mathcal{D},\ m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(6)\quad y_{m,t} \leq R^{\mathrm{d}}_{m,t} \left(\sum_{o\in\mathcal{I}} R^{\mathrm{c}}_{o,m,t} \cdot p_{o,m,t} + \sum_{o\in\mathcal{D}} q_{o,m,t} \right) + s_{m,t}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(7)\quad s_{m,t} = s_{m,t-1} + \left(\sum_{o\in\mathcal{I}} R^{\mathrm{c}}_{o,m,t} \cdot p_{o,m,t-1} + \sum_{o\in\mathcal{D}} q_{o,m,t-1} - y_{m,t-1} \right), \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}$$

$$(8)\quad \underline{X}^{\mathcal{K}}_t \leq \sum_{p\in\mathcal{K}} x_{p,t} \leq \overline{X}^{\mathcal{K}}_t, \quad \forall \mathcal{K} \in \mathcal{K},\ t \in \mathcal{T}$$

$$(9)\quad \sum_{p\in\mathcal{K}} \left(X_{p,0} + \sum_{\tau=1}^{t} x_{p,\tau} \right) \geq G^{\mathcal{K}}_t, \quad \forall \mathcal{K} \in \mathcal{K},\ t \in \mathcal{T}$$

$$(10)\quad p_{o,m,t},\ q_{o,m,t},\ s_{m,t},\ x_{p,t},\ y_{m,t},\ z_{c,t} \geq 0, \quad \forall o \in \mathcal{I} \cup \mathcal{D},\ m \in \mathcal{M},\ p \in \mathcal{P},\ c \in \mathcal{C},\ t \in \mathcal{T}$$

[7] Yao, B., & Dvorkin, Y. (2025, July). A decision framework for clean energy supply chain optimization. In Proceedings of the IEEE Power & Energy Society General Meeting (PESGM). IEEE. (Forthcoming)
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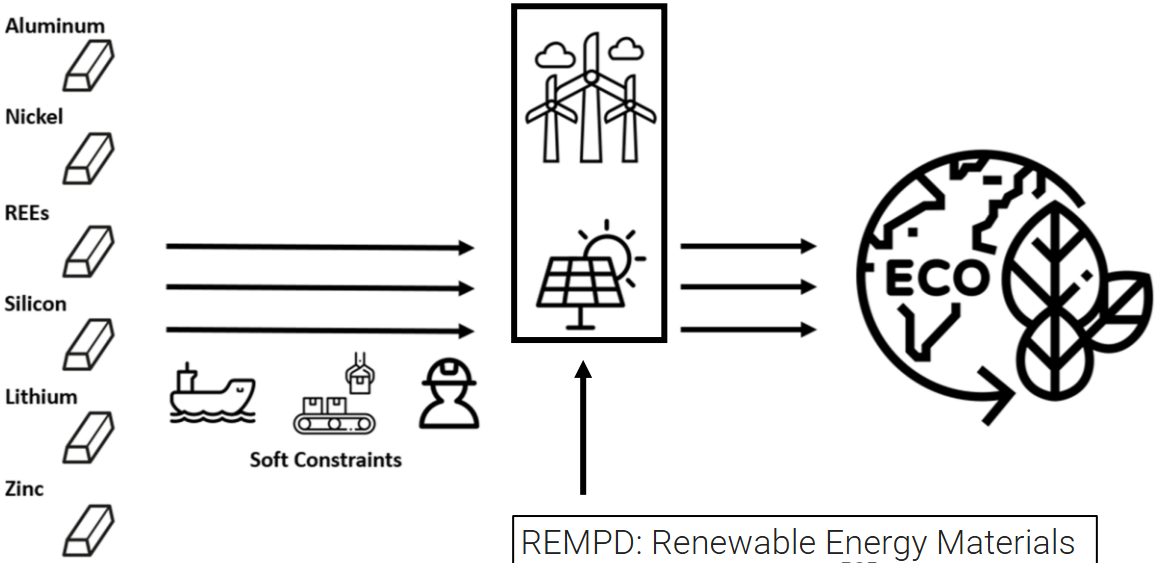

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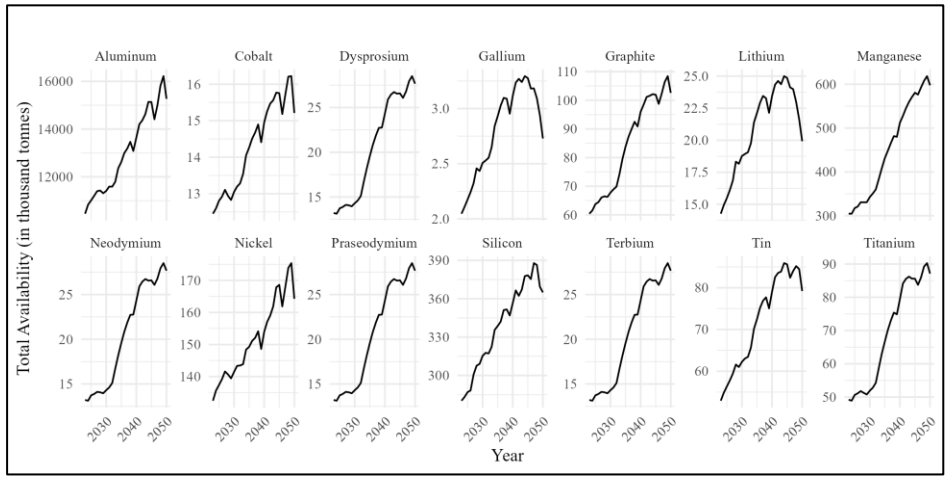


Objective = Minimize total costs for material procurement, component production, and product deployment.

Supply Chain: Sufficient materials → components → products.

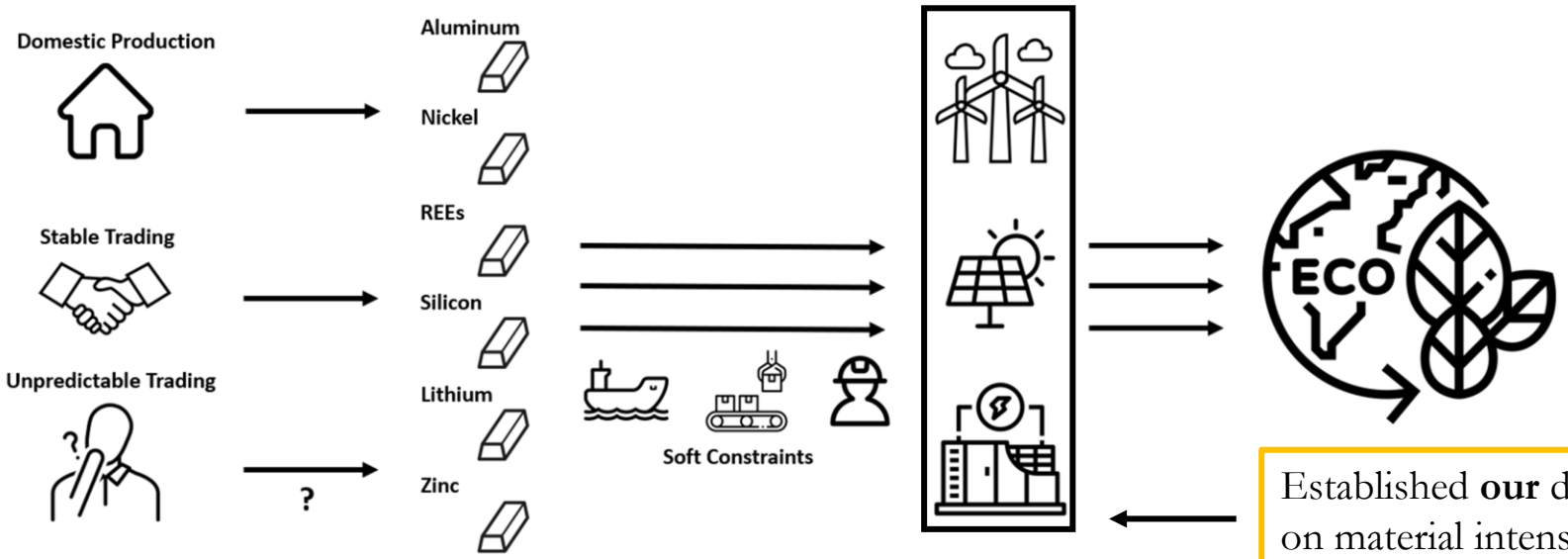
Material Acquisition: Procurement & production within limits; usage ≤ supply + stock.

Stock Management: Annual stock update from procurement, production, consumption.

Deployment Limits & Goals: Deployment within soft limits; meet cumulative goals (exogenous).

[7] Yao, B., & Dvorkin, Y. (2025, July). A decision framework for clean energy supply chain optimization. In Proceedings of the IEEE Power & Energy Society General Meeting (PESGM). IEEE. (Forthcoming)
[8] NREL (National Renewable Energy Laboratory). 2023. "Renewable Energy Materials Properties Database." Version 1.1. Washington, D.C.: U.S. Department of Energy. https://apps.openei.org/REMPD/.

Supply chain constraints for the US clean energy transition [9]



Material requirement for identified generation & storage resources.

+

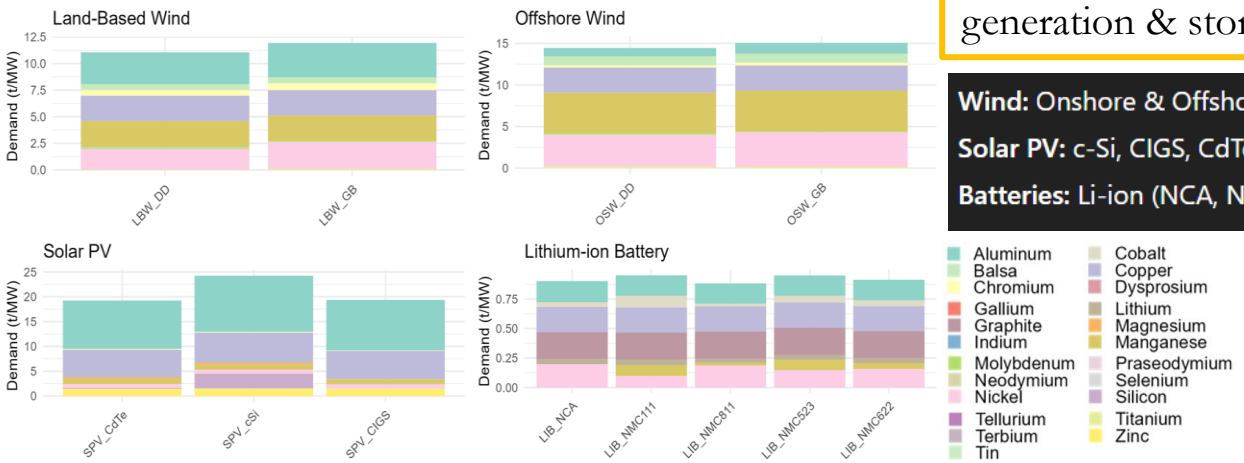
Projected future critical material availability.

+

Regulated capacity goals & soft limits (exogenous).

Established **our** dataset on material intensity for generation & storage.

14 → 26 Critical Materials;
Sources categorized as:
Stable vs. Unpredictable



Wind: Onshore & Offshore (Gearbox, Direct-drive)
Solar PV: c-Si, CIGS, CdTe
Batteries: Li-ion (NCA, NMC-111/523/622/811, LFP)

2050
Land-Based Wind – 400 GW Installation
Offshore Wind - 110 GW Installation
Solar Energy - 1600 GW Installation
Battery – 125 GW Installation

[9] Yao, B., Jeong, H., Mehrtash, M., Allan, B., Ockerman, D., & Dvorkin, Y. (2025). Understanding supply chain constraints for the US clean energy transition. npj Clean Energy, 1(1), 9.

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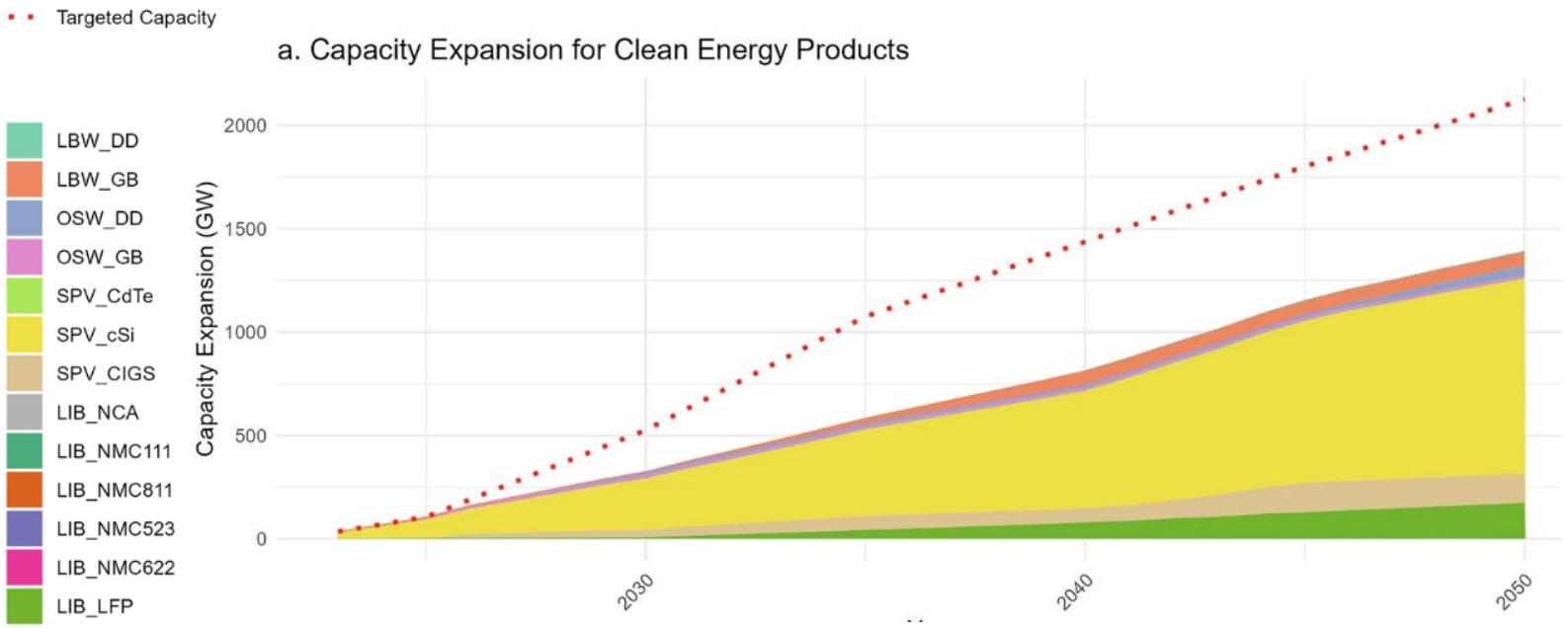

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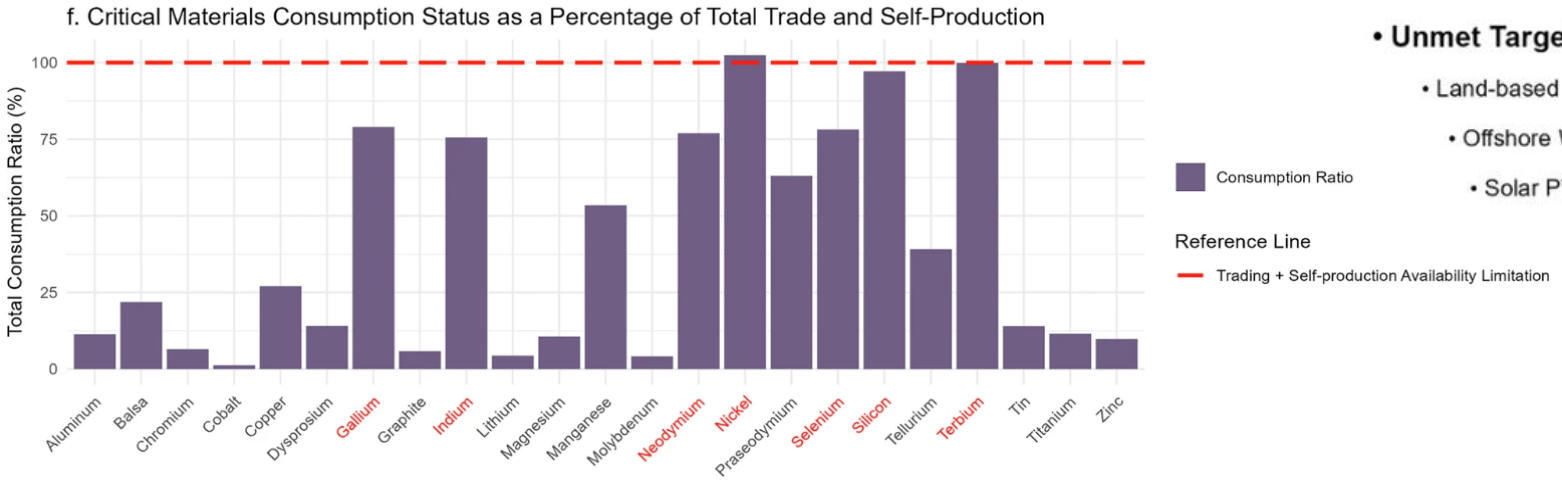


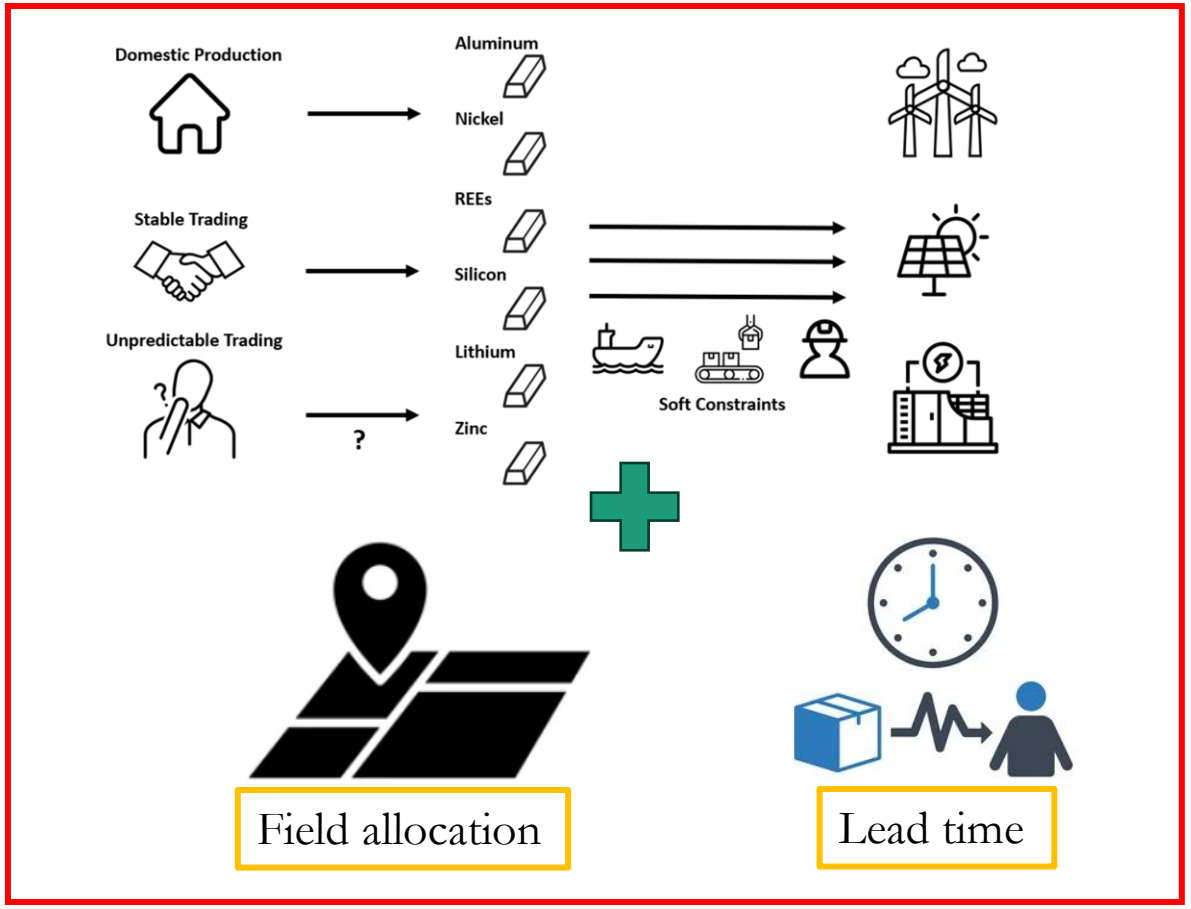
- **Unmet Targets 731.2 GW (34.4% of target)**
  - Land-based Wind 308.3 GW (81.2% of target)
  - Offshore Wind 48.4 GW (44.0% of target)
  - Solar PV 374.5 GW (25.6% of target)

Extra materials needed:

- **Neodymium:** 7,591 metric tons (+67.5%)
- **Nickel:** 1,355,659 metric tons (+106.2%)
- **Silicon:** 913,258 metric tons (+32.1%)
- **Tellurium:** 2,834 metric tons (+327%)

[9] Yao, B., Jeong, H., Mehrtash, M., Allan, B., Ockerman, D., & Dvorkin, Y. (2025). Understanding supply chain constraints for the US clean energy transition. npj Clean Energy, 1(1), 9.

But, more than just megawatts..



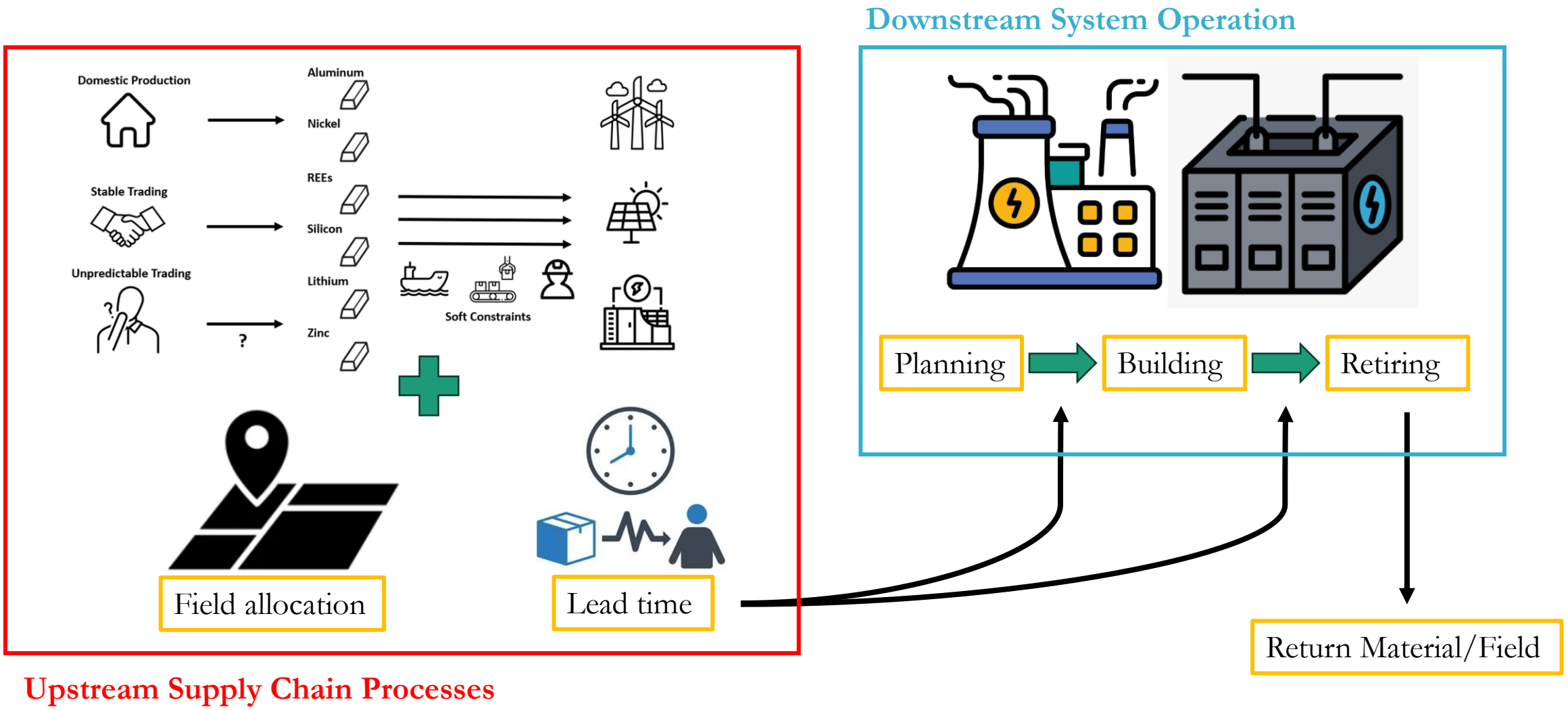
Upstream Supply Chain Processes

Regulated capacity goals (exogenous).

?

- Planning is not only about exogenous capacity goals.
- It must also account for real operational constraints.

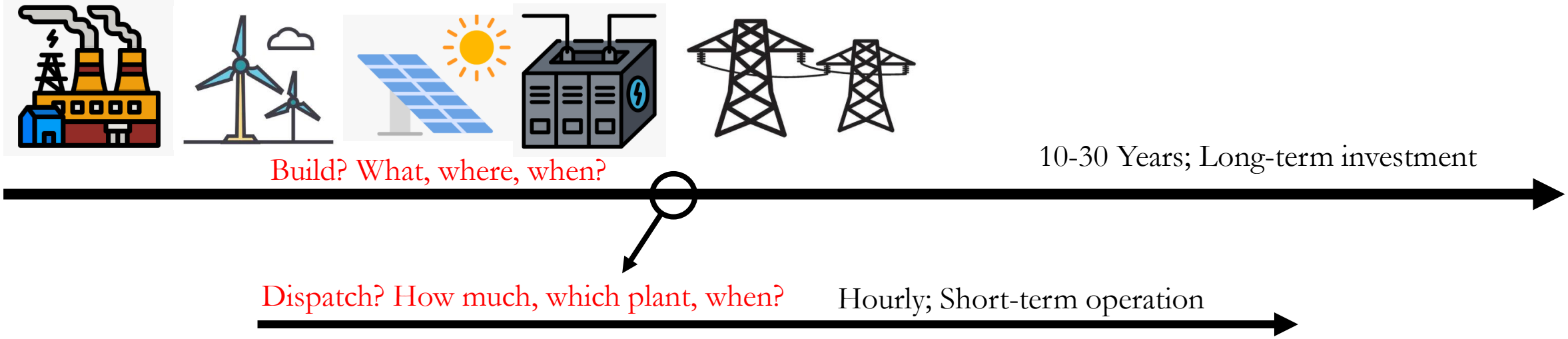
Integrating upstream supply chains into generation expansion planning [10]



[10] Yao, B., Bernstein, A., & Dvorkin, Y. (2025). Integrating Upstream Supply Chains into Generation Expansion Planning. arXiv preprint arXiv:2508.03001.

What is Capacity Expansion Planning

- **A long-term planning tool** decides what power plants, storage, and transmission to build, where, and when.
- **Balances trade-offs** among cost, reliability, and policy-driven goals under an uncertain future.
- **Integrates multiple time scales** i.e., short-term operations with long-term investment choices.
- **Informs policy and markets** by guiding regulators, utilities, and system operators on infrastructure and climate targets.



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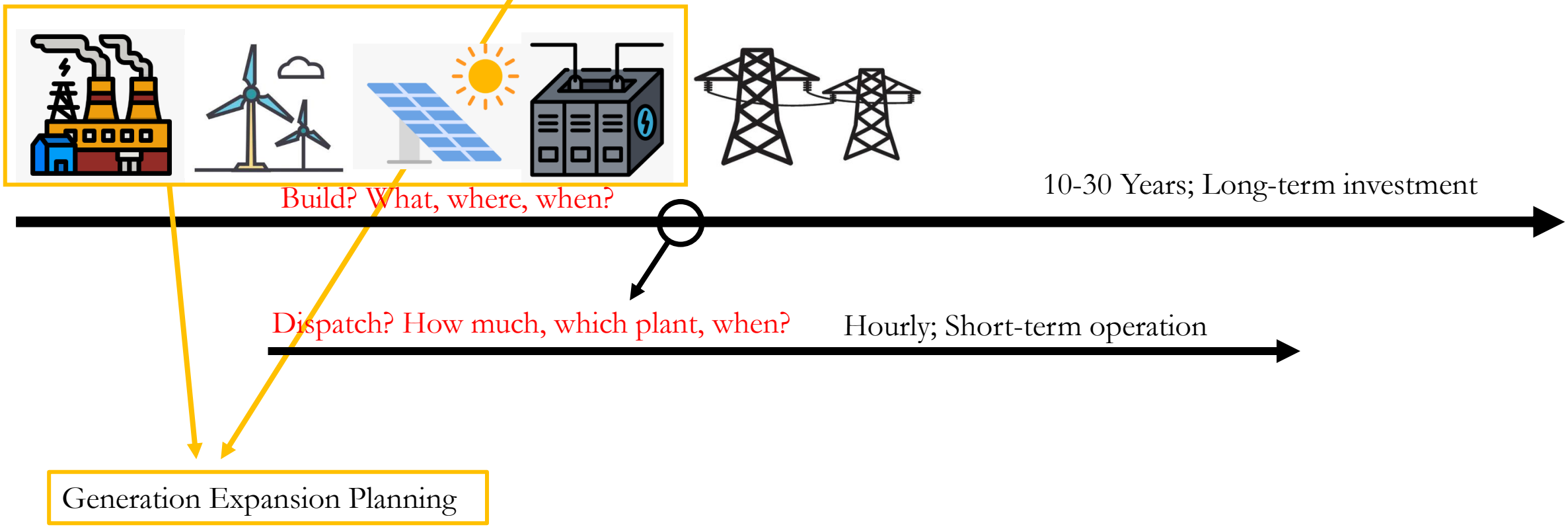
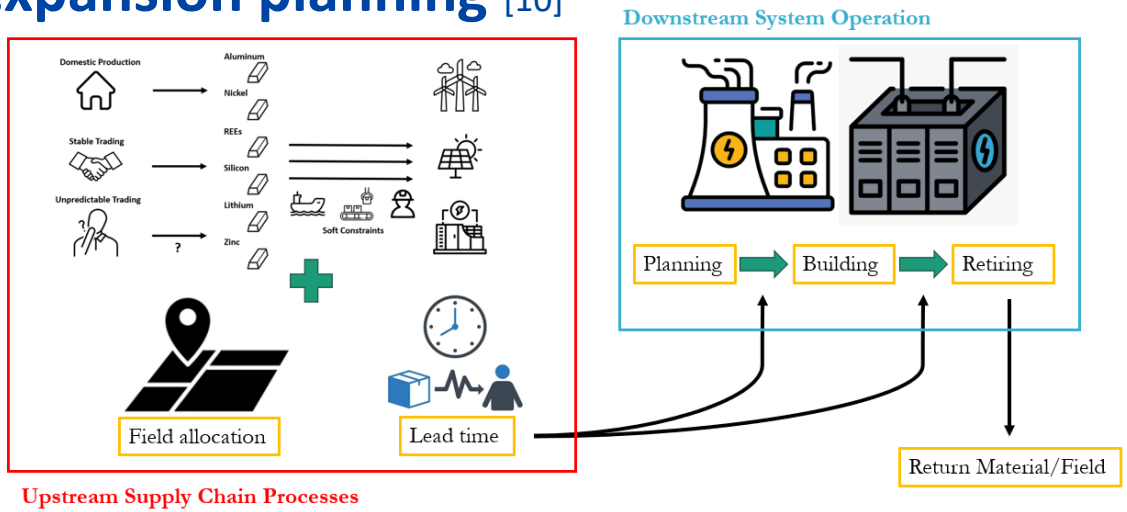


Integrating upstream supply chains into generation expansion planning [10]

- **Decisions (annual):** where/when/what to build or retire; product mix (e.g., SPV, LBW/OSW, BSS variants).
- **Operations (hourly, rep. days):** dispatch, storage charge/discharge, flows, load shedding.
- **Upstream layers:** materials → components → products mapped to capacity; lead time; stocks & recovery from retirements; field availability updates over time.



$$\min \sum_y C_y = \sum_y \left(C_y^{\text{in}} + C_y^{\text{op}} + C_y^{\text{pe}}\right) \quad (1a)$$

$$C_y^{\text{in}} = \sum_{g \in \bar{\mathcal{G}}} AC_{gy}^{\text{I}} \cdot \overline{P}_g^{\text{G}} \cdot d_{gy} \quad (1b)$$

$$C_y^{\text{op}} = \sum_g C_g^{\text{F}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy} + \sum_{g \in \mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h p_{gthy} + \sum_{g \in \mathcal{G}^{\text{st}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h (c_{gthy} + dc_{gthy}) \quad (1c)$$

$$C_y^{\text{pe}} = \sum_t N_{ty} \cdot P^{\text{VOLL}} \cdot \sum_h p_{ithy}^{\text{LS}} + P^{\text{RM}} \cdot p_y^{\text{RM}} + P^{\text{RPS}} \cdot e_{ky}^{\text{RPS}} \quad (1d)$$

$$u_{my} \geq \sum_c v_{cy} \cdot D_{mc}^{\text{CO}}, \quad \forall m, y \quad (2a)$$

$$v_{cy} \geq \sum_p w_{py} \cdot D_{cp}^{\text{PR}}, \quad \forall c, y \quad (2b)$$

$$u_{my} \leq \overline{M}_{my} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{gy} + s_{my}, \quad \forall m, y \quad (2c)$$

$$s_{my} = s_{m(y-1)} + \overline{M}_{m(y-1)} - u_{m(y-1)} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{g(y-1)}, \quad \forall m, y \quad (2d)$$

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot d_{gy} \leq \sum_{p \in \mathcal{P}^k} w_{py}, \quad \forall k, y \quad (2e)$$

$$\frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{gy}}{R_k^{\text{CAP}}} \leq f_{iy}^k, \quad \forall i, y, k \quad (2f)$$

$$f_{iy}^k = f_{i(y-1)}^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{gy}}{R_k^{\text{CAP}}} - \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{g(y-1)}}{R_k^{\text{CAP}}}, \quad \forall i, y > 1, k \quad (2g)$$

$$f_{i1}^k = A_i^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{g1}}{R_k^{\text{CAP}}}, \quad \forall i, k \quad (2h)$$

$$\sum_{y'' \leq y - T_g^{\text{LEAD}}} d_{gy''} = \sum_{y' \leq y} b_{gy'}, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2i)$$

$$b_{gy} = 0, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2j)$$

$$\sum_{y'' \leq y - T_g^{\text{LT}}} b_{gy''} = \sum_{y' \leq y} r_{gy'}, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2k)$$

$$r_{gy} = 1 \text{ if } y = T_g^{\text{RT}}, \ 0 \text{ if } y < T_g^{\text{RT}}, \quad \forall g \in \bar{\mathcal{G}}, \ \forall y \quad (2l)$$

$$\sum_{g \in \mathcal{G}_i \cap (\mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}})} p_{gthy} + \sum_{g \in \mathcal{G}_i \cap \mathcal{G}^{\text{st}}} (dc_{gthy} - c_{gthy}) - \sum_{l \in \mathcal{LS}_i} q_{lthy} + \sum_{l \in \mathcal{LR}_i} q_{lthy} = L_{ithy} - p_{ithy}^{\text{LS}}, \quad \forall i, t, h, y \quad (3a)$$

$$0 \leq p_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{th}}, \ y, t, h \quad (3b)$$

$$0 \leq p_{gthy} \leq F_{igthy}^{\text{GEN}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}_i \cap \mathcal{G}^{\text{rn}}, \ i, y, t, h \quad (3c)$$

$$o_{gy} = o_{g(y-1)} + b_{gy} - r_{gy}, \quad \forall g \in \mathcal{G}, \ y \geq 1 \quad (3d)$$

$$o_{g0} = 1 \text{ if } g \in \bar{\mathcal{G}}, \ 0 \text{ otherwise}, \quad \forall g \in \mathcal{G} \quad (3e)$$

$$-\overline{P}_l^{\text{L}} \leq q_{lthy} \leq \overline{P}_l^{\text{L}}, \quad \forall l, y, t, h \quad (3f)$$

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot F_{ky}^{\text{ELCC}} \cdot o_{gy} + p_y^{\text{RM}} \geq (1 + R_y^{\text{RM}}) \cdot \overline{L}_y, \quad \forall y, k \quad (3g)$$

$$0 \leq p_{ithy}^{\text{LS}} \leq L_{ithy}, \quad \forall i, y, t, h \quad (3h)$$

$$\sum_{g \in \mathcal{G}^k} \sum_t \sum_h N_{ty} \cdot p_{gthy} + e_{ky}^{\text{RPS}} \geq R_{ky}^{\text{RPS}} \cdot \sum_t \sum_h \sum_i N_{ty} \cdot L_{ithy}, \quad \forall k, y \quad (3i)$$

$$0 \leq c_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4a)$$

$$0 \leq dc_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4b)$$

$$0 \leq e_{gthy}^{\text{soc}} \leq \overline{E}_g^{\text{ST}}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4c)$$

$$e_{gthy}^{\text{soc}} = e_{gt(h-1)y}^{\text{soc}} + \epsilon^{\text{CH}} \cdot c_{gthy} - \frac{dc_{gthy}}{\epsilon^{\text{DC}}}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4d)$$

$$e_{gt1y}^{\text{soc}} = e_{gtendy}^{\text{soc}} = 0.5 \cdot \overline{E}_g^{\text{ST}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t \quad (4e)$$

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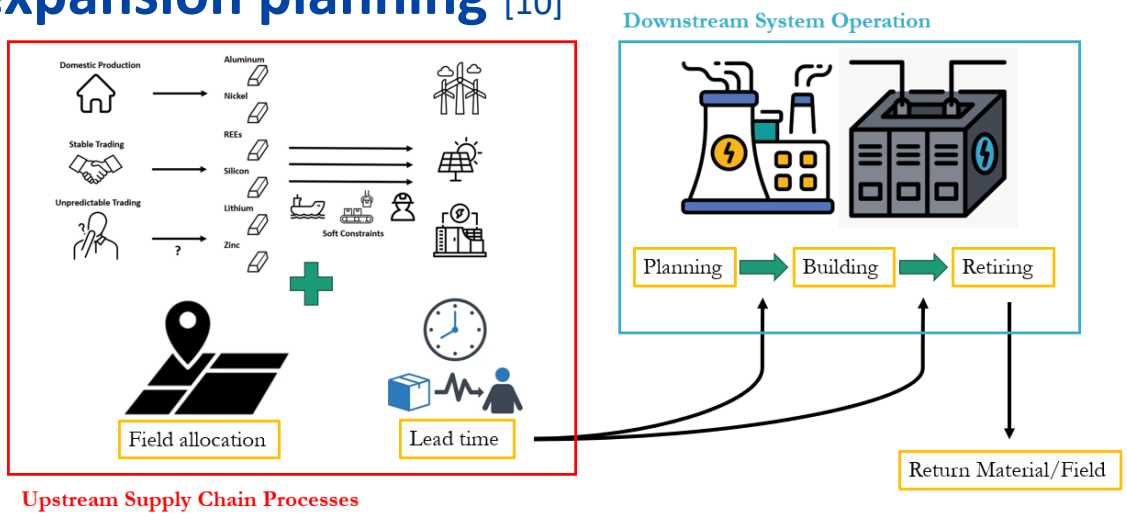

Objective = Investment + Operations + Penalties
Build new capacity, run the system, cover shortfalls.

Upstream layer:
1. Tracks materials → components → products.
2. Limits by production capacity & lead times.
3. Accounts for field availability.
4. Recovery from retired units.

Downstream layer:
1. **Balance**: generation + imports + storage = demand − shedding
2. **Limits**: generation, line capacity, load shedding bounds
3. **Evolution**: unit builds/retirements over years
4. **Battery:** charge/discharge ≤ rated capacity; SOC within bounds
5. **Reliability**: ELCC ≥ peak + reserves
6. **Compliance**: meet RPS targets (with slack)

[10] Yao, B., Bernstein, A., & Dvorkin, Y. (2025). Integrating Upstream Supply Chains into Generation Expansion Planning. arXiv preprint arXiv:2508.03001.

Integrating upstream supply chains into generation expansion planning [10]

- **Decisions (annual):** where/when/what to build or retire; product mix (e.g., SPV, LBW/OSW, BSS variants).
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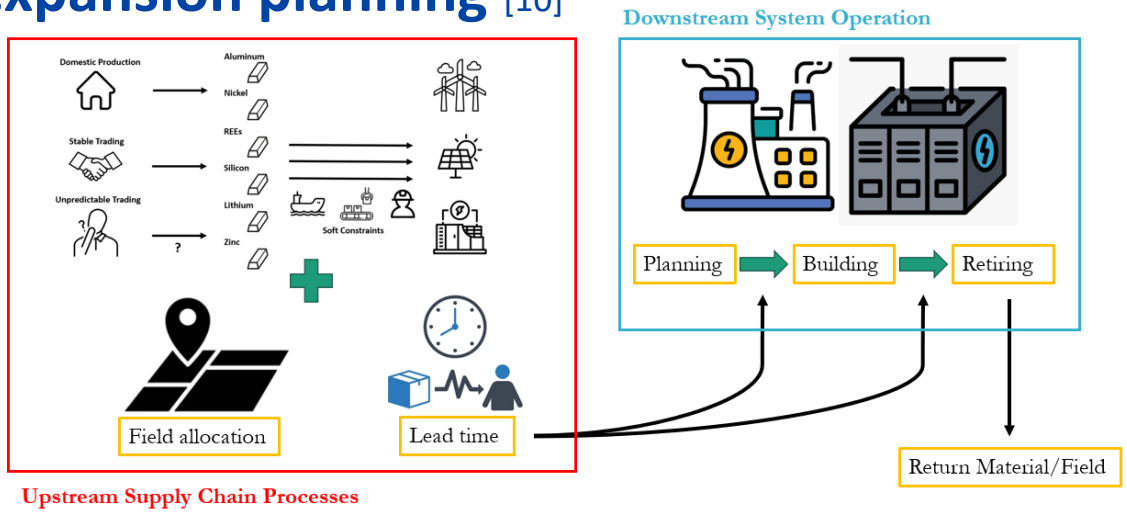

Material requirement for identified generation & storage resources.

+

Projected future critical material availability.

+

Regulated capacity goals & soft limits (exogenous).

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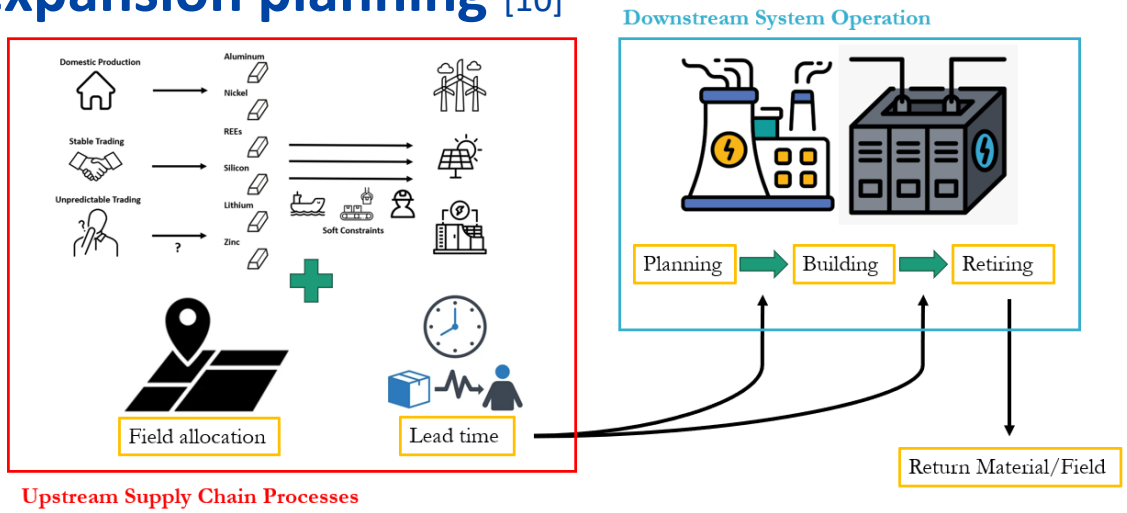


Material requirement for identified generation & storage resources.

+

Projected future critical material availability.

+

~~Regulated capacity goals & soft limits (exogenous).~~

+

Lead time & field availability.

→

Capacity & Technology Choices → Decision variables constrained by supply chain (endogenous).

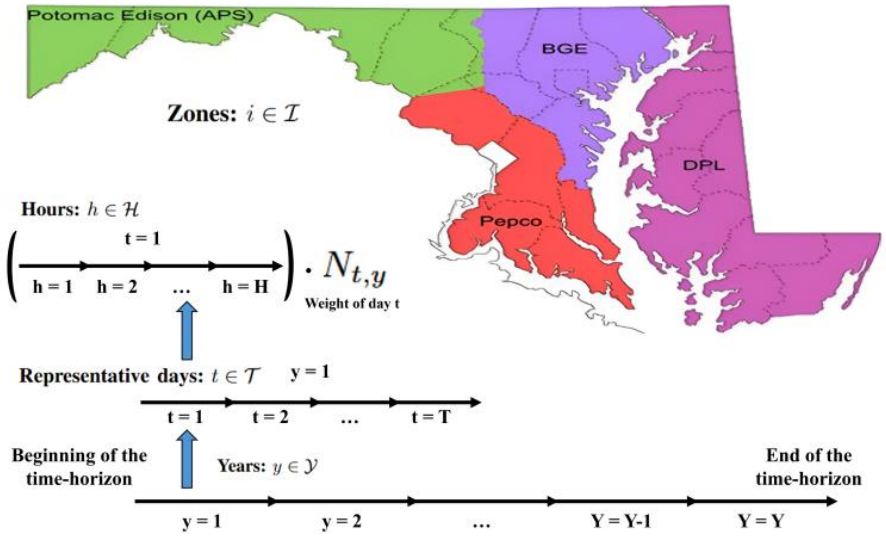
→

Capacity decisions → Ensure operational adequacy & reliability.

MILP → Accelerated by Benders Decomposition

[10] Yao, B., Bernstein, A., & Dvorkin, Y. (2025). Integrating Upstream Supply Chains into Generation Expansion Planning. arXiv preprint arXiv:2508.03001.

From Model → Maryland



Maryland spatial and temporal representation

- 4 zones (BGE, APS, DPL, PEPCO) as nodes.
- Investments made annually; operations modeled hourly with 1 representative day per season (via k-means clustering).
- Transmission: 4-node network, PJM Window 3 upgrades assumed, no further expansion.
- Focus: captures supply chain impacts in long-term planning.

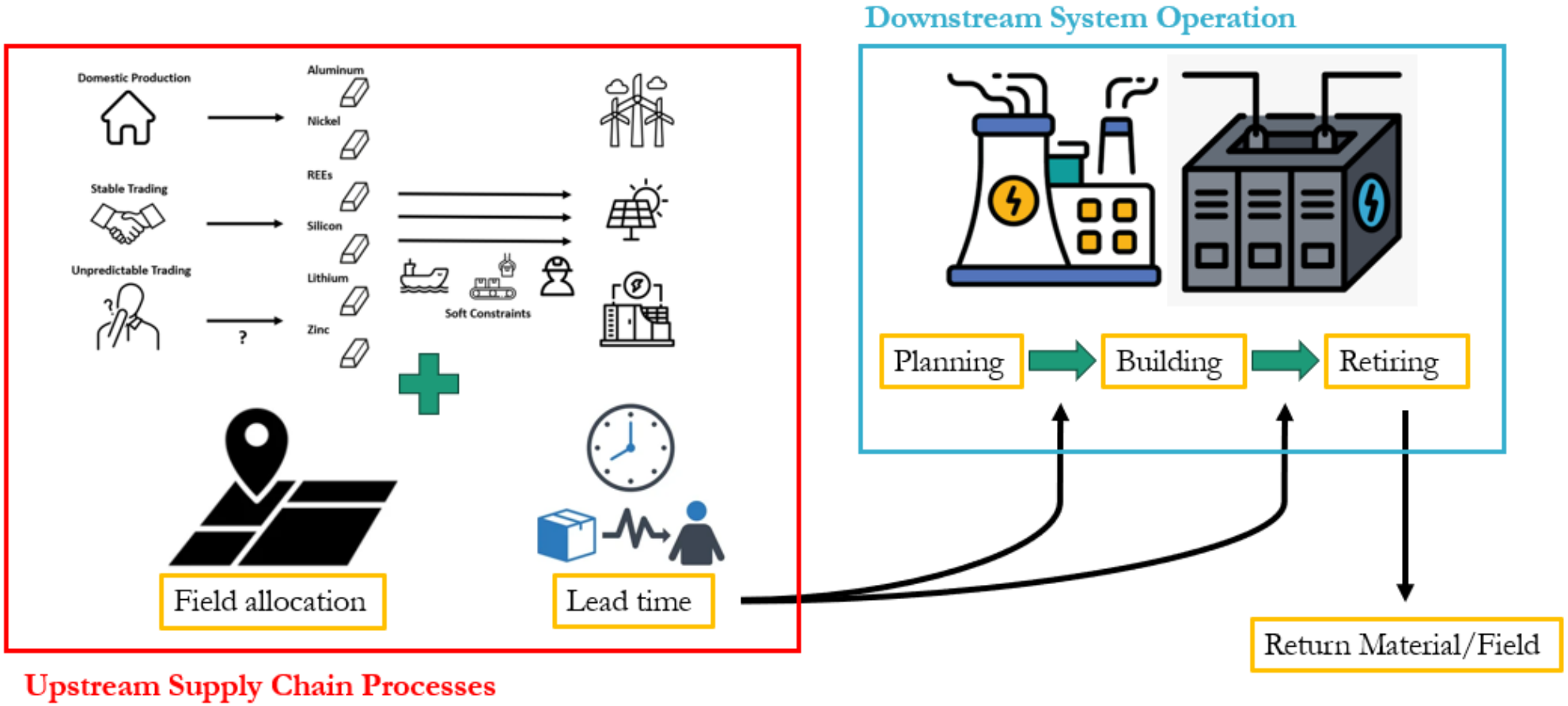
Assumptions

- Resources: existing + new (renewables, batteries only) (per MD regs & queue).
- Lifetimes: retire at design end (+1 year if exceeded).
- Materials: 14 critical materials → 12 key products.
- Supply: domestic, imports, stock, 10% recycling.
- Lead times: fixed tech-specific delays.
- Field: land & offshore limits; shared zones; reuse after retirements.

Scenarios:

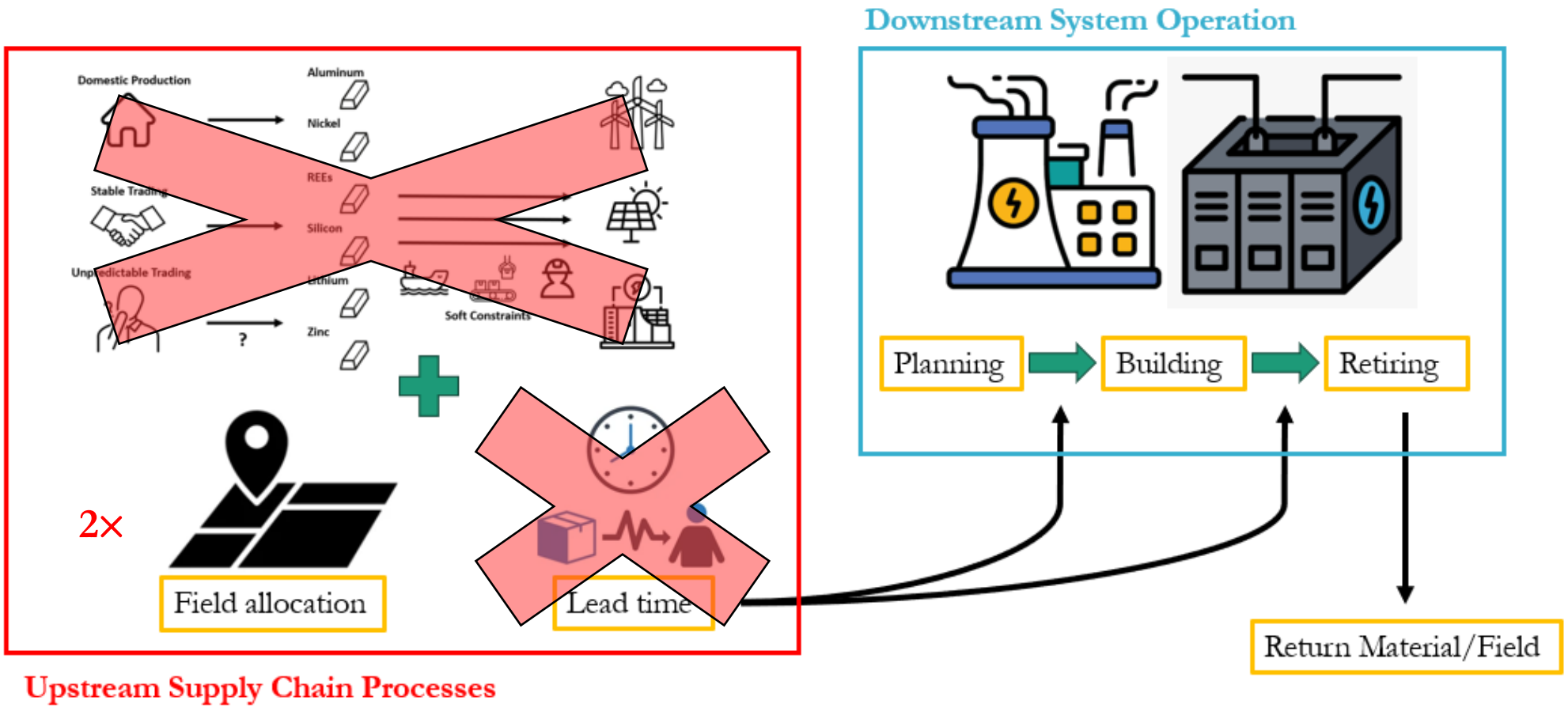
- **Low**: excludes aggressive electrification & data center growth.
- **High**: includes both electrification & data center growth.
- Sub-Scenarios:
  - **w/o SC** (relaxed): unlimited materials, no lead times, expanded land/offshore areas.
  - **lim. SC** (constrained): materials limited to domestic + allied sources, with lead times & land limits reflecting geopolitical/trade risks

From Model → Maryland



- **Baseline**

From Model → Maryland



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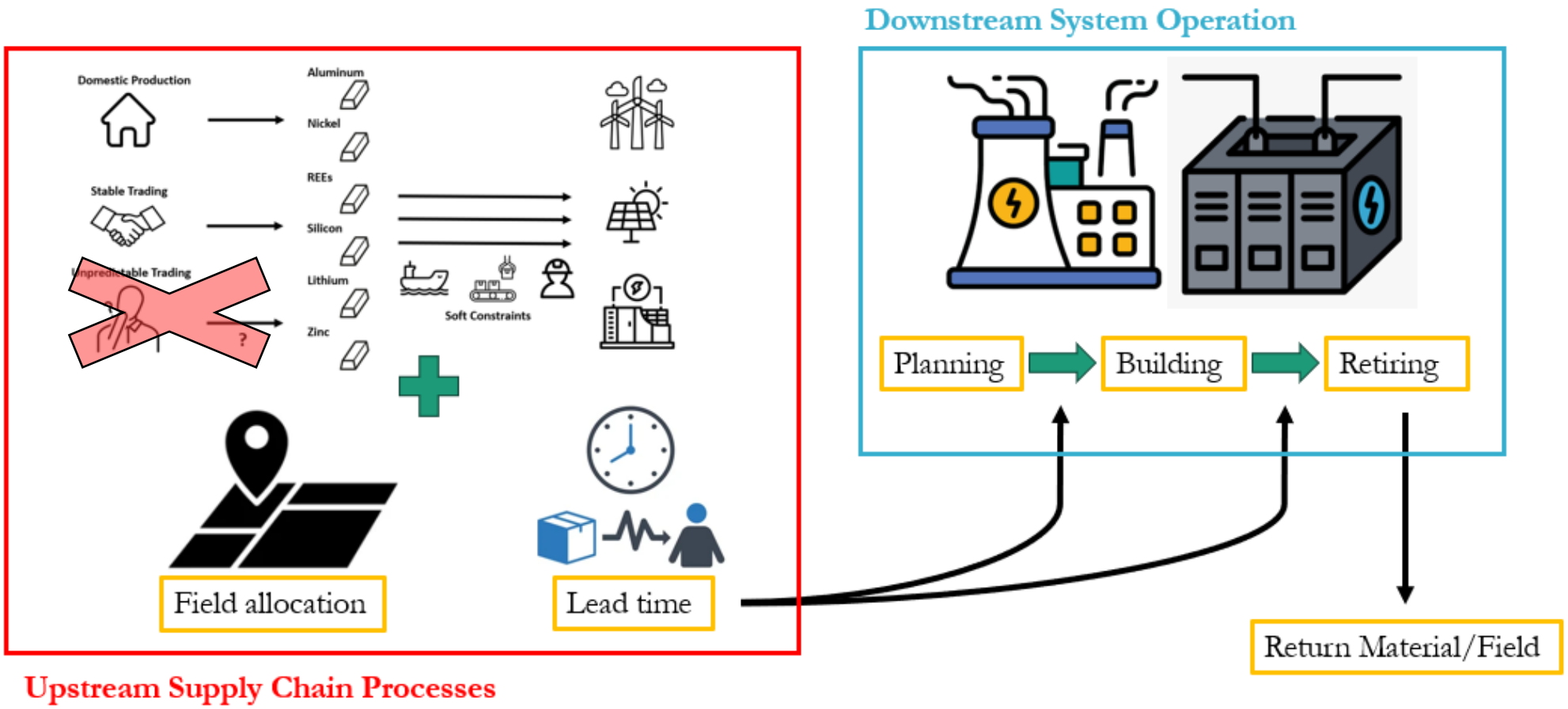
From Model → Maryland



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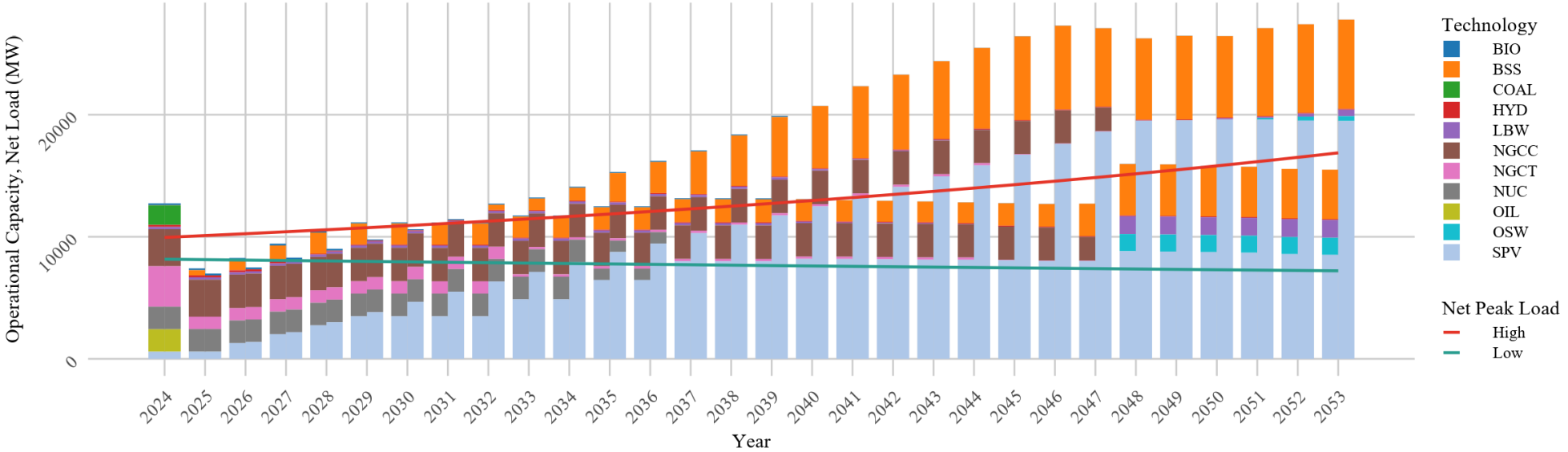
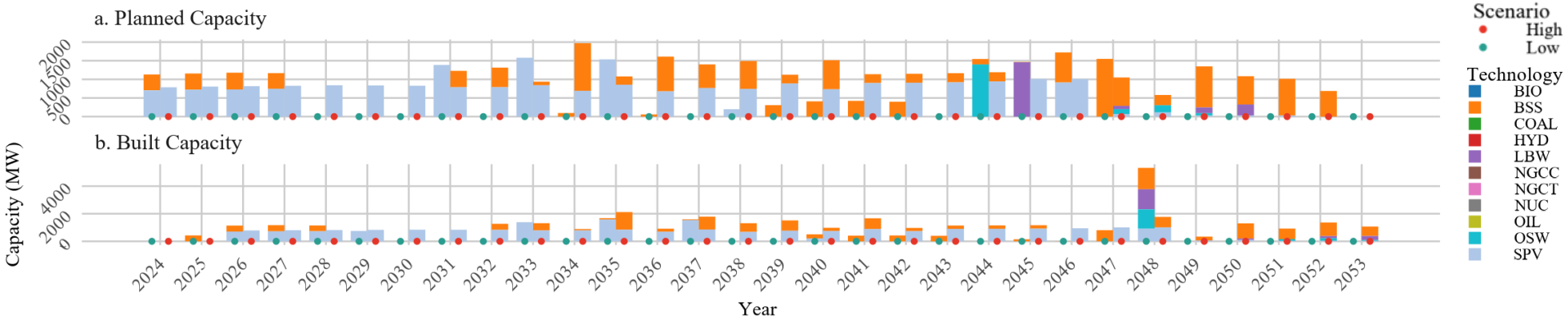


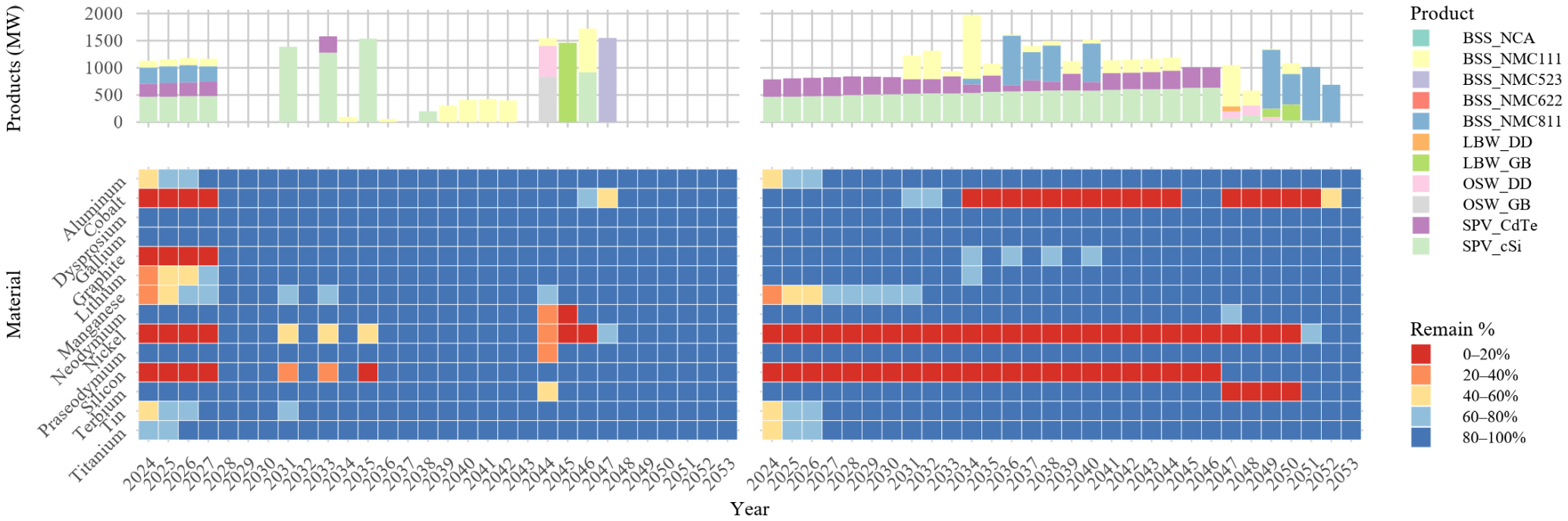
Operational capacity over time for baseline Low and High scenarios. Stacked bars: tech-specific capacity (Low baseline = left); Lines: net peak load (system demand)

- Low baseline (2053): 4.1 GW BSS, 1.5 GW LBW, 1.4 GW OSW, 8.5 GW SPV → more diversified mix.
- High baseline (2053): 7.3 GW BSS, 0.5 GW LBW, 0.4 GW OSW, 19.5 GW SPV → solar-dominant growth.



Optimization Results for Status Variables: (a) Planning Capacity, (b) Built Capacity.

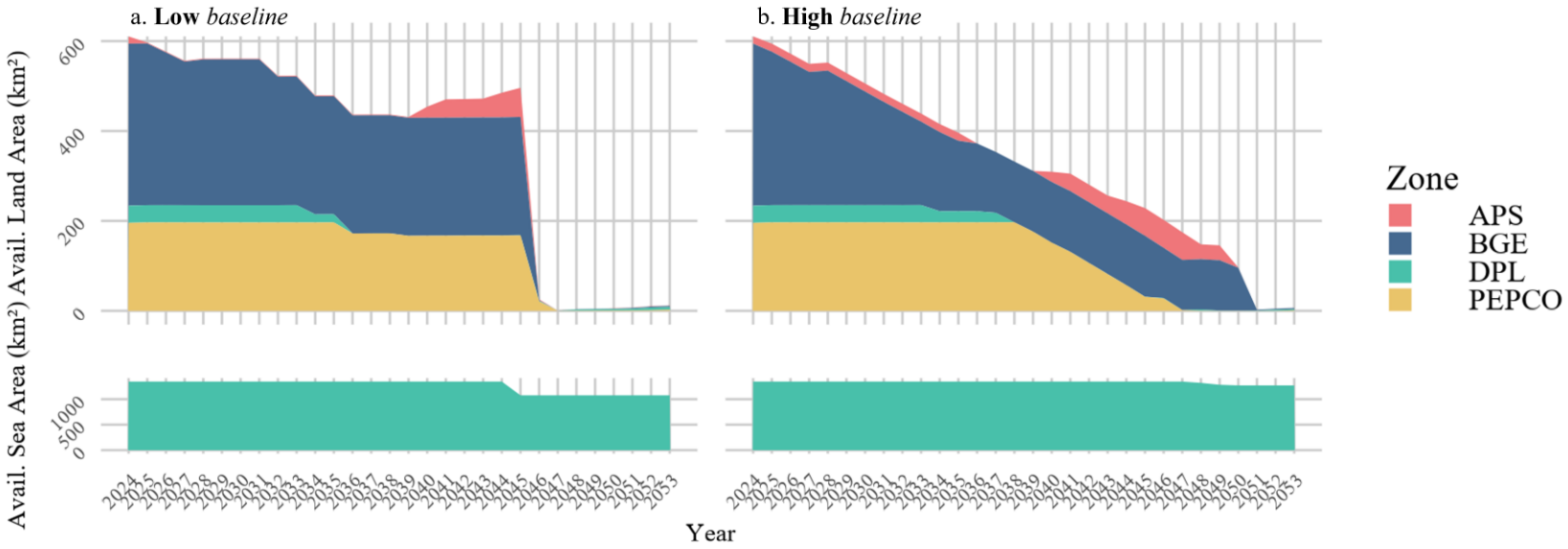
From Model → Maryland



Planned product by technology (top) and remaining material availability (bottom) for baseline Low and High scenarios.

- Low baseline:
  - Pre-2031: SPV/BSS constrained by silicon & nickel bottlenecks.
  - 2040s: shift to LBW & OSW as costs fall and materials accumulate.
  - Mix of gearbox & direct-drive OSW to balance nickel vs. neodymium use.
- High baseline:
  - Continuous load growth → sustained SPV build to 2046 (fast build, low material use).
  - Storage shifts from NMC 111 (cobalt) to NMC 811 (nickel).
  - 2047: SPV insufficient → add LBW & OSW (direct-drive favored for nickel limits).

From Model → Maryland



Field availability at the beginning of each year: land (top) and offshore (bottom) area for new deployment.

- Post-2049: land outside BGE fully allocated → spatial constraint binds
- Response: shift to BSS (low land need, short lead time)
- Materials: move toward NMC 811 (less cobalt) while keeping some NMC 111 for balance

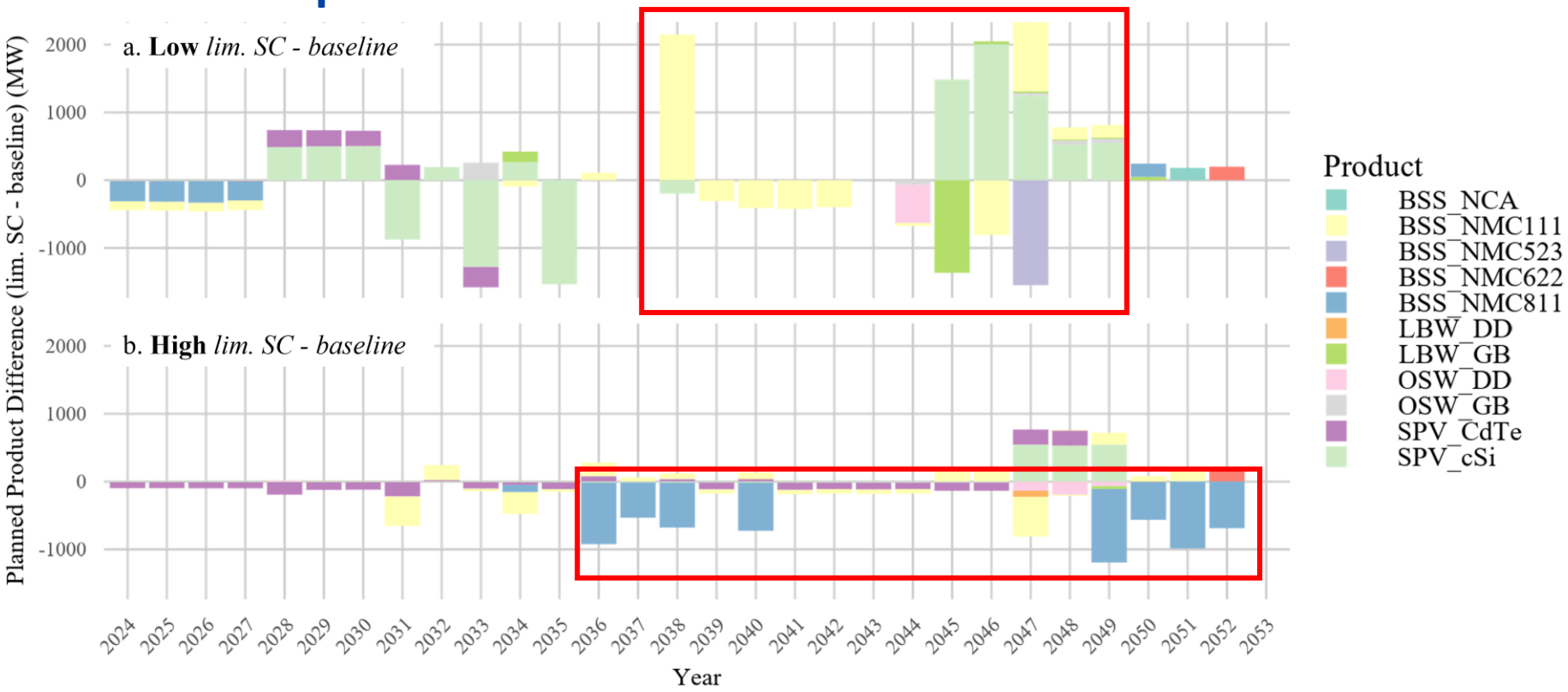
Scenario Comparisons

Without supply chain constraints, plans build only when needed.

Difference in planned capacity by technology between the w/o SC and baseline. Positive values indicate technologies favored in w/o SC.

- **w/o SC (relaxed):**
  - Unlimited materials & no lead times → just-in-time planning.
  - Low: LBW added in 2048 (cheap & land available).
  - High: sustained load growth → more OSW after 2045.

Scenario Comparisons

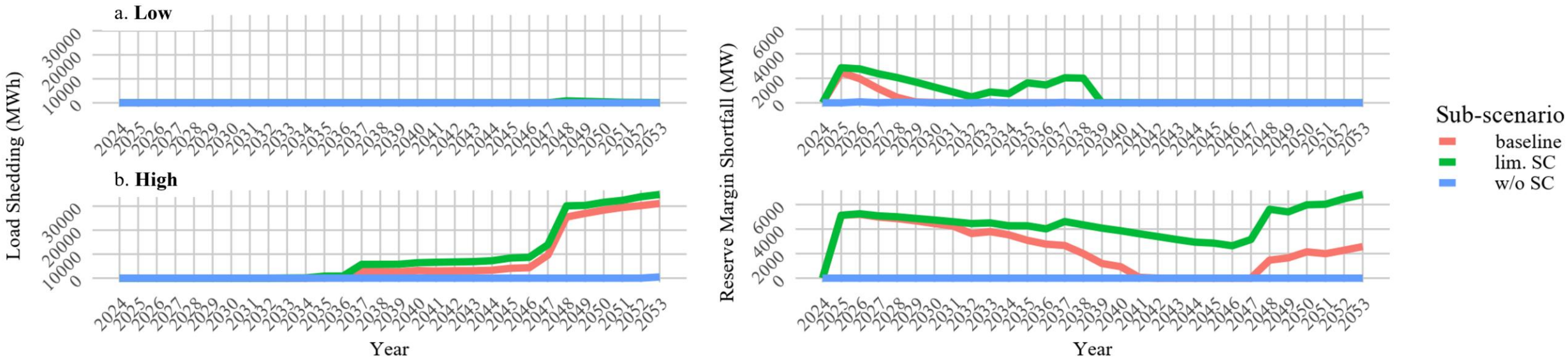


Difference in planned product deployment between the lim. SC and baseline. Positive values indicate technologies favored in lim. SC.

- c-Si preferred → strong domestic supply
- NMC-811 not preferred → nickel import dependence

- **lim. SC (constrained):**
  - Critical material limits (esp. rare earths) → no new wind after 2045.
  - SPV (c-Si) favored due to steady silicon supply.
  - BSS shift: NMC 811 → NMC 111 (less nickel).
  - Stronger shift in High case due to higher load stress.

Scenario Comparisons



Yearly load shedding and reserve margin shortfalls across all scenarios.

- **Reliability:**
  - w/o SC: almost no issues (flexible, fast build).
  - baseline: early shortfalls (2025 retirements), mostly resolved by 2041.
  - lim. SC: persistent reserve margin violations; load shedding in High from 2037 onward; major gap after 2048 NGCC retirements remains unresolved.

Key Takeaways

- Ignoring supply chains underestimates costs and reliability risks.
- Low vs. High: modest cost increase in Low; severe shortfalls and load shedding in High.
- Just-in-time planning is infeasible — requires earlier, strategic investments and possible lifetime extensions.
- Constraints can overturn cost advantages — diversification across technologies is critical.
- Anticipatory planning needed to avoid over-correction and material exhaustion.